The model, not the search

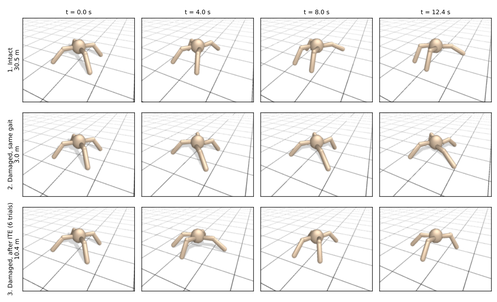
Few-trial damage recovery with Quality-Diversity repertoires

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The question

A robot that loses a leg should find a new gait in as few trials as possible: on hardware every trial costs time and wear. Intelligent Trial and Error (ITE, Cully et al., *Nature* 2015) learns a **repertoire** of diverse gaits in simulation with Quality-Diversity (QD), then, after the damage, uses Bayesian optimisation to choose which gaits to try. Modern QD algorithms build repertoires with reinforcement learning (RL) and are evaluated with an oracle that tries every gait.

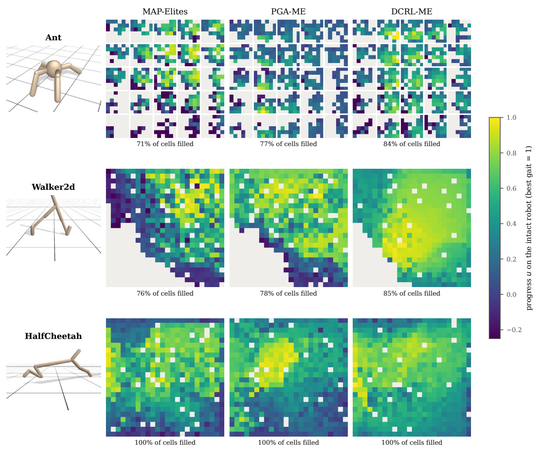
How many trials does recovery really take, and what limits it?



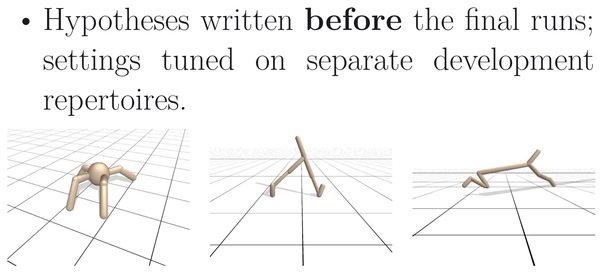
Ant: best gait, same gait with a leg paralysed, gait found by ITE in six trials.

Setup

- **3 robots** (Brax): Ant, Walker2d, HalfCheetah. **33 repertoires** of 1024 cells built by MAP-Elites, PGA-ME and DCRL-ME at equal budget.
- **Damages**: limbs paralysed or weakened (expected family); motor stuck or wired backwards, lost sensors, slippery or heavy foot (unexpected); a simulated reality gap.
- **Replay**: every gait played under every condition; 15 methods compared over 20 repetitions; validated against fresh episodes (correlation 0.97–0.99).
- Hypotheses written **before** the final runs; settings tuned on separate development repertoires.



One repertoire per robot and algorithm: each square is a gait placed by its foot contacts, coloured by its progress on the intact robot.



1 Repertoires survive damage

54 / 73 / 64 %
of the intact progress still reachable after damage on Ant (MAP-Elites / PGA-ME / DCRL-ME); no significant difference

5–13×
better gaits left after damage in repertoires built with RL

5
median trials for ITE, against hundreds of episodes to re-train with TD3

2 ITE barely beats a ranking

Trying the best intact gaits in order (**top-k**) is almost as good as ITE: 27 % against 28 % of the intact progress after 3 trials on Ant. A descriptor that models the damage three times better does not help, nor does a reality gap; on Walker2d the descriptor kernel fails entirely.

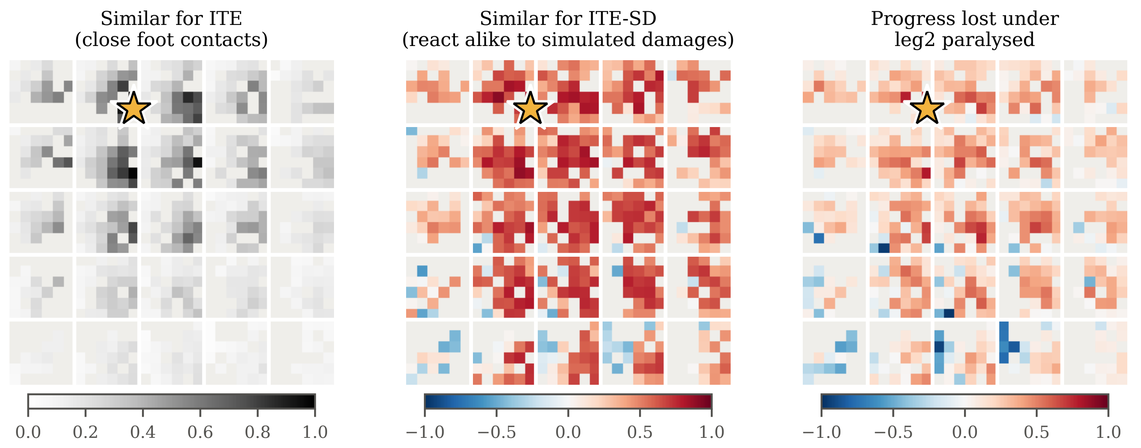
Diagnosis

The **model** of ITE, not its search, is the bottleneck: its prior ignores which gaits survive damages, and its similarity, closeness of foot contacts, was chosen for diversity, not to predict how gaits react to a damage.

3 Learning how gaits react to damage

Before deployment, the simulator breaks the robot **64 times**. Playing every gait under each simulated damage gives the response matrix $R_{ki} = u_i(\delta_k) - p_i$, at no cost on the robot.

- **ITE-SD**: prior $m_i = p_i + \bar{r}_i$ and kernel $k_{\mathrm{SD}}(i,j) = \mathrm{cov}_k(R_{ki}, R_{kj}) + \tau^2 \mathbb{1}[i = j]$, the empirical-Bayes prior of Wang et al. (NeurIPS 2018) learned here on the gaits of a repertoire. The GP becomes a Bayesian linear model of the damage: each trial identifies the damage and predicts every gait.
- **ITE-Sig**: each trial also returns a proprioceptive signature (joints, torso, foot contacts), explained by the same latent damage. One trial informs the model through many numbers instead of one.



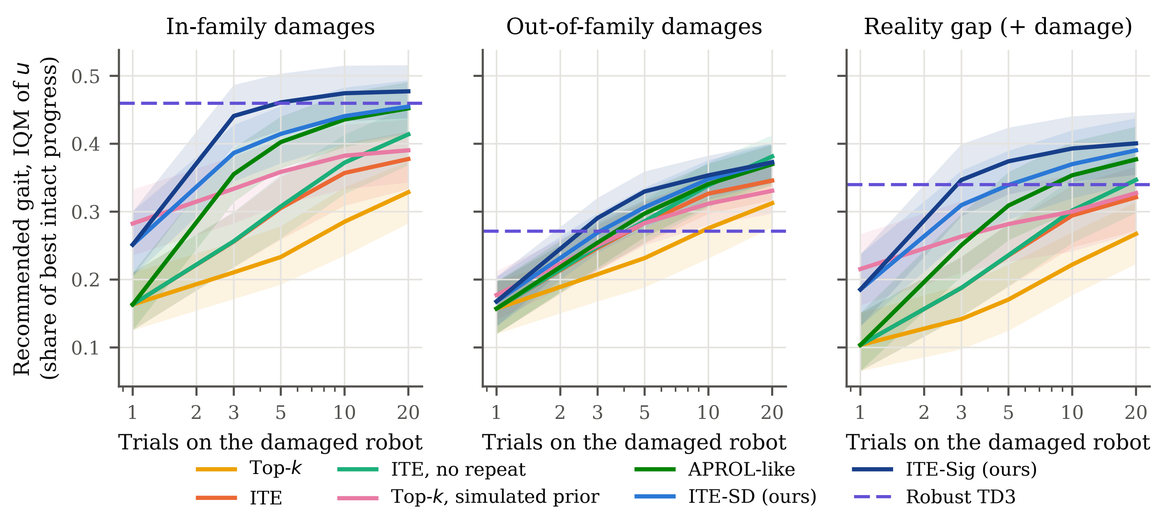
Gaits similar to the best intact gait (star) for ITE (left) and for ITE-SD (middle), and the progress actually lost when a leg is paralysed (right, red: loses). The star is the same gait in the three panels.

4 Results

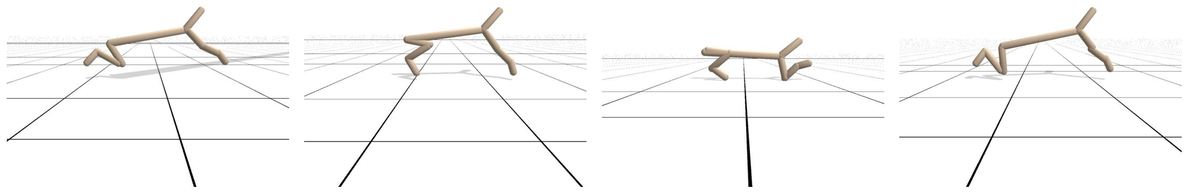
0.31 → 0.41
share of the intact progress after 5 trials, ITE → ITE-SD (expected damages, 3 robots)

0.46
with ITE-Sig, better than ITE-SD in every condition family

3×3
robots and QD algorithms on which ITE-SD beats ITE ($p \leq 0.001$)



- Gains in the expected family (+0.079 vs ITE) and under a reality gap (+0.072); **not significant on unexpected damages** (+0.010).
- A model helps most when the intact ranking misleads (Spearman −0.38 between gain and intact/damaged rank correlation).
- On Ant and HalfCheetah the robot **really goes further**: the chosen gaits cover 45 % → 55 % (Ant) and 49 % → 57 % (HalfCheetah) of the intact distance, ITE → ITE-Sig.
- On Walker2d the score rewards standing: the learned model picks gaits that **do not fall** (47 % vs 20 % for ITE) rather than gaits that walk further.



HalfCheetah, back leg weakened: intact gait, same gait damaged, after 5 trials of ITE, after 5 trials of ITE-Sig (videos on the project page).

5 Repertoires or reinforcement learning?

- **Re-training** with TD3 needs hundreds of episodes on the damaged robot (median 448 with a critic warm-up), against 1 to 20 trials for a repertoire.
- A **damage-robust policy** trained on random damages needs no trial and beats every adaptation on Ant within its family (0.84 vs 0.48), but loses its advantage on unexpected damages (0.34 vs 0.32). Combining both is the natural next step.

Take-home

- Modern QD repertoires stay useful after damage.
- With its original model, Bayesian search adds little to a ranking.
- Learning the model from a few dozen simulated damages makes